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JOURNAL OF
Creative Behavior
Publication of the Creative Education Foundation™

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"Who" Is the Best Creative Thinking Partner? An Experimental Investigation of Human–Human, Human–Internet, and Human–AI Co-Creation

ABSTRACT

Recent research suggests that working with generative artificial intelligence (AI), such as ChatGPT, can produce more creative outcomes than humans alone. However, does AI retain its creative edge when humans have access to alternative information sources, such as another human or the internet. We explored this question in a between-group experiment with 202 German participants across four conditions (human–human dyads, human–Internet, and two human–AI groups with basic or specific instructions) and four creativity tasks (two alternate uses tasks, a consequences task, and a problem-solving task). Results showed that the human–human condition obtained higher creativity scores in the divergent thinking tasks than the remaining groups. No significant between-group differences were observed in the problem-solving task. Moreover, interacting in human dyads made people more creatively confident, an effect not observed in the other groups. In addition, we compared human-rated outcomes with AI-based automated scoring (Ocsai). Interestingly, notable discrepancies emerged between the AI assessment and the human-judged results, raising concerns about AI's susceptibility to "elaboration bias." These findings highlight the benefits of human collaboration for creativity and call for further studies about the reliability and potential biases of AI in evaluating creative performance.

Keywords: generative AI, ChatGPT, cognitive enhancement, creative confidence, divergent thinking, AI scoring systems, elaboration bias.

With recent advances in artificial intelligence (AI) technology, AI has tapped into various fundamental human abilities, such as creativity. This expansion has made AI a hot topic in creativity research, inspiring many to explore AI's different roles in the creative process (e.g., Garcia, 2024; Holmström & Carroll, 2024; O'Toole & Horvát, 2024; Sueur, Lombard, Capra, Beltzung, & Pelé, 2024). Most of the research on AI and creativity has focused on two forms of AI application: First, whether AI has the potential to enhance human creativity either as a tool or an equal collaborator in the creative process (e.g., Eapen, Finkenstadt, Folk, & Venkataswamy, 2023; Habib, Vogel, Anli, & Thorne, 2024; Jia, Luo, Fang, & Liao, 2024; Urban et al., 2024), and second, whether AI can be more creative than humans, potentially replacing us (e.g., Cropley, 2023; Guzik, Byrge, & Gilde, 2023; Haase & Hanel, 2023; Hubert, Awa, & Zabelina, 2024).

The first AI application scenario has several important practical implications, enabling people to enhance their creativity and collaborate with AI on creative tasks at work or in artistic endeavors. Many researchers suggest that the most promising use of AI, especially generative AI, is the augmentation of human creativity, enabling people to cocreate with the technology (e.g., Elfa & Dawood, 2023; Ivcevic & Grandinetti, 2024; Vinchon et al., 2023). Indeed, a recent analysis reported that idea generation is one of the most common use cases for generative AI (Zao-Sanders, 2024). Through the widespread integration of generative AI tools like ChatGPT into our daily lives (see Duarte, 2024), the question arises whether creative outputs generated through the collaborative efforts of humans and AI (human–AI cocreativity) will emerge as the new gold standard, gradually replacing human creativity. This makes the comparison of human creativity and human–

AI cocreativity a new important area of research (see Rafner, Beaty, Kaufman, Lubart, & Sherson, 2023). Yet, what do we know so far about human–AI cocreativity?

HUMAN–AI CREATIVE COLLABORATION: PROCESSES AND OUTCOMES

In contrast to individual creativity, collaborative creativity and cocreation involve a substantial amount of interaction and coordination among collaborators (Paulus, Dzindolet, & Kohn, 2012). In human-to-human cocreation, while exchanging ideas, cognitive (e.g., combining ideas), social (e.g., discussing viewpoints), and motivational (e.g., competing) processes enrich and stimulate the cocreation process (Paulus, 2000; Paulus et al., 2012). Moreover, as creative collaboration usually develops around a complex task, the task itself, its perceived difficulty, and commitment to it influence creative collaboration (Paulus et al., 2012; Reiter-Palmon, Wigert, & de Vreede, 2012). Although cocreation does not always lead to more creative output, the additional stimulation and activation of new knowledge can outperform individuals who do not interact with others in creative idea generation (Nijstad, 2015), for example, through variation and recombination of available knowledge—a core aspect of creativity (Finke, Ward, & Smith, 1992; Weisberg, 1993).

These mechanisms also seem to apply to human–AI cocreation, especially with generative AI, where the creative process is stimulated through collaboration. Generative AI is a type of AI technology that generates content such as texts or images extrapolated from an extensive database (Feuerriegel, Hartmann, Janiesch, & Zschech, 2024). In this way, the creative process is enriched by a large source of information and knowledge provided by the AI, for example, by interacting and exchanging ideas with the AI in different ways (e.g., AI being an advisor) (Rezwana & Maher, 2023) or by using the AI's knowledge to think more broadly and explore possible variations of creative ideas more interactively (O'Toole & Horvát, 2024).

Recent studies have demonstrated that cocreation efforts with generative AI, such as ChatGPT, can indeed enhance human creativity. For instance, generative AI has been shown to improve the idea generation and evaluation process, positively impacting the creative output of brainwriting sessions (Shaer, Cooper, Mokryn, Kun, & Shoshan, 2024). In another study, Habib et al. (2024) observed that students using ChatGPT in an alternative uses test (AUT) achieved higher creative test scores regarding originality, flexibility, and fluency than students who did not use ChatGPT. Similarly, Urban et al. (2024) found that students using ChatGPT showed higher originality and quality in their creative solutions, while also aligning more closely with task goals in completing creative problem-solving tasks than their peers who did not use the AI tool. Moreover, students who used ChatGPT perceived the task as easier and requiring less mental effort.

However, there is reason to believe that human–AI cocreativity has to be more rigorously studied and compared to other cocreation processes, rather than simply juxtaposed against individual human creativity. For example, a field experiment by Dell'Acqua et al. (2023) compared the performance of different groups of consultants working on creative product development, with some groups using ChatGPT and others not. The study showed that although the consultants using ChatGPT produced higher quality ideas, using ChatGPT also reduced the variation in the ideas produced, making them more similar to each other and jeopardizing idea originality. In contrast, consultants working without ChatGPT produced more diverse and conceptually varied ideas. Similarly, a study comparing story writing showed that AI-assisted stories were more similar to each other than stories written by humans alone (Doshi & Hauser, 2024). This observation was echoed by a study that found that human-crowdsourced solutions outperformed ChatGPT-generated solutions in novelty, even when their quality was comparable (Boussioux, Lane, Zhang, Jacimovic, & Lakhani, 2024). The decline in novelty when cocreating with generative AI has also been demonstrated in the arts (Zhou & Lee, 2024).

There is no doubt that using generative AI and its vast knowledge base enriches the collaboration process for creativity (see O'Toole & Horvát, 2024). Thus, indeed, human–AI cocreation can be superior to human-only conditions. However, as the research described earlier demonstrated, this evidence often oversees the comparability between different groups in the cocreation process. Because generative AI provides external cognitive support, research should also compare whether any other sources of external knowledge can produce creative outputs similar to those of human–AI cocreativity. This would help us understand if non-AI external cognitive support enriches creativity in the same way as AI. For example, what would happen if humans used the Internet or other people's perspectives as sources of knowledge in their creative thinking process? Would this be as augmentative as using AI? Or perhaps even more so? Answering this question is essential for both theory and practice because, as described earlier, it would shed light on whether human–AI cocreation is truly superior to other forms of cocreation. Therefore, we posit that it is

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premature to conclude that AI solutions consistently outperform humans in creative tasks; quite the opposite: there are still aspects to be empirically explored to provide a better understanding of the boundary conditions and nuances of the differences between people working together and working with AI while creating. This is precisely the gap this study aims to address.

ARE PEOPLE MORE CONFIDENT WITH AI BY THEIR SIDE?

Beyond the effects of human–AI collaboration on creative outcomes or the perceived cognitive support provided by AI tools, interacting with AI is also likely to shape participants' perceptions of their own creativity. For instance, despite improved idea generation when cocreating with AI, people might feel concerned that the machine is taking over the thinking process (Habib et al., 2024), potentially diminishing their creative confidence and engagement. Similarly, McGuire, De Cremer, and Van de Cruys (2024) showed that the level of influence people had in the cocreation process with AI affected their creative self-efficacy or confidence. People who had more control over the cocreation process with AI reported higher levels of creative confidence, while people who had less control over the creative process reported lower levels of creative confidence.

Conversely, if people perceive a task as easier and more manageable because of AI support, it can make them feel more creatively empowered and thus improve their creative self-perception. For example, as shown by Urban et al. (2024), students who used ChatGPT not only performed better on creative problem-solving tasks, but also perceived these tasks as less challenging than their peers who worked alone. Importantly, working with ChatGPT resulted in participants' higher perceived ability to solve the creative task (i.e., task-specific creative self-efficacy) compared to a working-alone condition. This higher task-specific and dynamic creative self-efficacy may have vital implications, given the growing literature showing that so-called agency factors such as creative self-beliefs (Karwowski, Lebuda, & Beghetto, 2019) and self-regulation (Zielińska, Forthmann, Lebuda, & Karwowski, 2023) make effective dealing with creative tasks more likely.

Yet, while it might seem evident that AI solutions strengthen people's creative confidence, as Urban et al. (2024) demonstrated in the case of dynamic creative self-efficacy, it is not always the case. A recent study (Faiella, Zielinska, Karwowski, & Corazza, 2024) showed that people's AI-related creative self-beliefs are generally lower than their general creative self-beliefs. In other words, participants in this study did not feel more creative when using AI, at least compared to what they typically thought about their own creative abilities. Given the importance of creative agency, also in the AI context (Beghetto & Karwowski, in press; Faiella et al., 2024), this investigation digs deeper into how working with people versus working with machines (AI, Internet) might influence individuals' creative confidence.

THIS STUDY

So far, research has largely overlooked the comparability of different cocreation partners with human–AI cocreativity. This study addresses this gap by employing an experimental design that allows us to investigate whether the creative outputs of human–AI cocreation differ from other forms of collaboration in terms of creativity and quality. By treating AI as a cognitive enhancement that provides information, we examine whether other forms of cocreation utilizing different external cognitive enhancements produce similar creative outcomes. For this purpose, we set up four experimental groups, each working with a different cognitive enhancement: generative AI (i.e., ChatGPT), the Internet (i.e., Google Search), or another human. The Human–ChatGPT group was further divided into two subgroups with different instructions on how to use ChatGPT.

The main research questions this study addresses are: (RQ1) How do participants evaluate their cocreation processes with different cocreation partners (i.e., their cognitive enhancements) in terms of task interest, task difficulty, mental efforts, and partner's contribution? (RQ2) How does working on the creativity tasks with different cocreation partners affect participants' creative confidence? (RQ3) Do the different interactions result in varied creativity levels of the generated ideas across the four groups?

While our study primarily focused on the characteristics of the idea-generation process and outcomes when people collaborate with machines versus humans, our experimental settings also offer a unique opportunity to contribute to the discourse on using AI for idea scoring. Despite human ratings being considered the gold standard for creativity judgments (Long & Plucker, 2015; Plucker & Makel, 2010), AI-based automated ratings show promise in not only effectively mimicking these scores (Organisciak, Acar, Dumas, & Berthiaume, 2023; Patterson, Barbot, Lloyd-Cox, & Beaty, 2024), but also offering more objectivity in some

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respects (e.g., Dumas, Dong, Kagan, & Campbell, 2024). However, this form of automated scoring has only been introduced recently, and despite its promising performance in previous investigations (e.g., DiStefano, Patterson, & Beaty, 2024; Johnson et al., 2022; Luchini et al., 2023), as well as the creation of a new creativity test for children entirely based on automated scoring (Acar, Dumas, Organisciak, & Berthiaume, 2024), much remains to be understood about its strengths and challenges. To explore this further, we replicated our analyses using AI-based automated scoring.[1] Could AI models trained to mimic human scores "prefer" ideas generated with ChatGPT? Alternatively, might these models rate such ideas lower because they can predict what ChatGPT might generate? Although these questions are not central to our research goals, addressing them contributes to the broader discussion on integrating AI in creativity research by testing the validity of automated scoring methods within experimental contexts.

METHOD

PARTICIPANTS

A total of 210 students from a German university participated in our study. Eight non-native German speakers were excluded from data analysis to reduce possible language-induced variability in verbal creativity tasks. Following this data screening strategy, the final sample comprised 202 participants (99% German natives), with 43.1% females. The participants' ages ranged from 17 to 51 years, with a mean age of 22.74 years (SD = 4.00). The majority (77.7%) were bachelor's degree candidates, and the rest were pursuing master's degrees, all within the field of business management. Specifically, 43.6% were majoring in Business Administration, 31.2% in Sport Management, 17.3% in Business Psychology, and 7.9% in Business Informatics.

Table 1 displays the sociodemographic characteristics of the participants, illustrating comparability across the four groups in terms of age, $F(3, 184) = 1.50$, $p = .217$; gender distribution, $\chi^2(6) = 6.11$, $p = .412$; study areas $\chi^2(9) = 4.90$, $p = .843$; and study levels, $\chi^2(3) = 4.22$, $p = .239$. With regard to participants' prior experience with ChatGPT, 11.4% of participants reported no prior usage of ChatGPT, though they might have heard about it, 59.4% had used ChatGPT once or occasionally, and 29.2% identified themselves as regular users, engaging with it weekly or daily. There were no significant differences in prior experience with ChatGPT across the groups, $\chi^2(18) = 20.91$, $p = .284$. Similarly, the groups did not differ in their openness to experience, $F(3, 198) = 0.79$, $p = .503$, a vital predictor of creativity (Puryear, Kettler, & Rinn, 2017).

PROCEDURE

The experiment was structured in three parts (see Table 2). In Part 1, the students completed a short survey about their background (gender, age, study area, and study level), prior experience with ChatGPT, and the self-assessments of their openness to experience and creative confidence. In Part 2, they collaborated with their human or digital partners (Google Search/ChatGPT) on four creativity tasks. Upon completion of these tasks, a subsequent questionnaire was administered in Part 3 to examine their prior experience with similar tasks and to assess the levels of task interest, task difficulty, and mental efforts related to the tasks. Furthermore, this survey also evaluated the perceived helpfulness of their assigned partners. The final section of the survey included a second measurement of participants' creative confidence.

Participants were recruited via email announcements and face-to-face briefings on the university campus. The experiment was conducted across three laboratories, where students interacted with one peer student (Lab 1), Google Search (Lab 2), or ChatGPT (Lab 3). In Lab 1, student dyads were paired randomly and worked in separate rooms to prevent disturbances. Lab 2 was configured to allow participants access exclusively to Google Search, with all other search engines and AI tools being blocked. Similarly, Lab 3 was set up to allow access only to ChatGPT 3.5, restricting the use of any other search engines or AI resources. After being randomly assigned to one of the four conditions, students received the same instructions in the same sequence, outlining that they were going to complete creativity tasks with a partner (either a peer student, Google Search, or ChatGPT) and were encouraged to use these partners as a source of inspiration or knowledge while leveraging their own creativity to fulfill the tasks. Students completed creativity tasks with their cognitive support in a two-step approach, which we designed based on the extensive literature that underscores the significance of both divergent and convergent thinking in creative cognition (Benedek & Fink, 2019; Eysenck, 2003; Finke et al., 1992). In line with meta-analyses by Said-Metwaly, Fernández-

[1] We thank an anonymous reviewer for inspiring this line of analyses.

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TABLE 1. Sociodemographic Characteristics of the Participants

	Human –human (n = 41 dyads)	Human –Internet (n = 39)	Human–ChatGPT (basic instruction) (n = 39)	Human–ChatGPT (specific instruction) (n = 42)	Full sample (n = 202)
Gender					
Female	38 (46.3%)	16 (43.2%)	18 (46.2%)	15 (35.7%)	87 (43.1%)
Male	43 (52.5%)	21 (51.4%)	21 (53.8%)	24 (57.1%)	109 (53.9%)
Divers	0 (0.0%)	1 (2.6%)	0 (0.0%)	0 (0.0%)	1 (0.5%)
Missing	1 (1.2%)	1 (2.6%)	0 (0.0%)	3 (7.2%)	5 (2.5%)
Age					
Min.	18	18	18	17	17
Max.	51	29	35	29	51
M	23.39	22.69	22.46	21.74	22.74
SD	4.95	2.89	3.56	2.63	4.00
Study area					
BusPsy	18 (21.9%)	5 (12.8%)	6 (15.4%)	6 (14.3%)	35 (17.3%)
BusAdm	34 (41.5%)	17 (43.6%)	19 (48.7%)	18 (42.9%)	88 (43.6%)
BusInf	8 (9.8%)	3 (7.7%)	1 (2.6%)	4 (9.5%)	16 (7.9%)
SportM	22 (26.8%)	14 (35.9%)	13 (33.3%)	14 (33.3%)	63 (31.2%)
Study level					
Bachelor	58 (70.7%)	31 (79.5%)	33 (84.6%)	35 (83.3%)	157 (77.7%)
Master	24 (29.3%)	8 (20.5%)	6 (15.4%)	7 (16.7%)	45 (22.3%)
Nationality					
German	41 (A) (100%) 39 (B) (95.1%)	39 (100%)	39 (100%)	42 (100%)	200 (99.0%)
Openness (1–5 scale)					
M	3.67	3.80	3.76	3.66	3.69
SD	0.78	0.79	0.72	0.96	0.80

Note. BusPsy = Business Psychology, BusAdm = Business Administration, BusInf = Business Informatics, SportM = Sport Management. (A) = Person A and (B) = Person B in Group 1. Person A assumed a leadership role by guiding Person B through the process and directly engaging with the creativity tasks. This includes explaining task requirements to Person B, soliciting their support throughout the tasks, and ultimately providing the responses. Person B, in contrast, adopted a supportive and passive role, assisting Person A without actively completing any of the creativity tasks.

Castilla, Kyndt, and Van den Noortgate (2020) on the impact of time limits and instructions on creative performance, our experiment did not impose any time restrictions. This decision was based on evidence suggesting that explicit instructions, such as those employed in our study, significantly enhance task originality without time constraints. Most participants spent about 35 min (SD = 11.47) to complete the experiment. Ethical approval was secured from the Ethical Committee of the participating university, adhering to American Psychological Association (APA) ethical principles. Participants were informed of the objectives of the study and the measures taken to protect their data. Formal consent was obtained from all participants before they started the study.

MEASURES

Openness to experience was measured using five items from the short German Big Five Inventory (BFI-K) (Rammstedt & John, 2005). Responses were given on a 5-point Likert scale ranging from 1 (*not at all true to me*) to 5 (*completely true to me*) on items like "I have a wide range of interests." This scale demonstrates good internal consistency across the four groups, with Cronbach α ranging from .74 to .83, and .78 for the whole sample.

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TABLE 2. An Overview of the Materials and Instruments Used in the Study

Part 1: Pretest survey	Part 2: Creative thinking tasks	Part 3: Posttest survey
Sociodemographics (gender, age, study area, study level) Openness to experience (five items) (Rammstedt & John, 2005) Creative confidence (six items) (Karwowski, Lebuda, & Wisniewska, 2018) Prior experience with ChatGPT (one item) (Urban et al., 2024)	Task 1: AUT of pants Task 2: AUT of a fork Step 1: *List at least 10 unusual uses of pants/a fork (divergent thinking)* Step 2: *Select the top 3 ideas and further elaborate them.* Task 3: Consequences task Step 1: *List at least 10 possible consequences if people no longer needed or wanted any food.* Task 4: Creative problem-solving Step 2: *Select the most serious consequence of task 3 and suggest a solution that is both original and useful (convergent thinking).* Participants were instructed to finish all tasks with the help of their peers (Google Search/ ChatGPT).	Prior experience with creative thinking tasks (1 item) Perception of the: • *task interest (one item)* • *task difficulty (one item)* • *mental effort (one item)* (Urban et al., 2024) Evaluation of the • *helpfulness of their peers (Google Search/ChatGPT) in (1) generating new ideas; (2) further developing the ideas; (3) solving the problems* • *contributions of the participants and their partners in solving the creative tasks.* Creative confidence (six items) (Karwowski et al., 2018)

Note. AUT = alternate uses task.

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Creative confidence was measured with six items of the Short Scale of Creative Self (Karwowski et al., 2018). Participants rated how well each statement described them on a 5-point Likert ranging from 1 (*definitely not*) to 5 (*definitely yes*), with items such as "I think I am a creative person." The measure showed high internal consistency across the four groups, with Cronbach α ranging from .69 to .86, averaging .80 for the whole sample in the pretest. For the posttest, values ranged from .79 to .89, with an average of .86 for the entire sample.

Perceived task interest, task difficulty, and mental effort were measured using the items from Urban et al. (2024), which were based on Schiefele (2009) and Silvia (2005) for perceived task interest, Efklides (2006) for perceived task difficulty, and Paas (1992) for mental effort. The original 100-point scale was adapted to a 101-point scale, ranging from 0% (*not at all interesting/difficult; no effort at all*) to 100% (*extremely interesting/difficult; all my efforts*). This adaption was made to facilitate participant responses and allows for nuanced measurement of the variables.

Helpfulness of the partner was assessed with three self-developed items (Cronbach's α = .92), examining how helpful the participants perceived their partner (human peers, Google Search, ChatGPT) in (1) generating new ideas, (2) further developing the ideas, and (3) solving the problems. Participants responded to a 101-point scale, ranging from 0% (*not at all helpful*) to 100% (*extremely helpful*).

Contribution of oneself and the partner was evaluated with two self-developed items, which asked the participants to estimate the extent of their own and their partners' contributions in solving the creative tasks, quantified from 0% to 100%. Participants' estimates were not required to sum up to 100%.

Creative thinking tasks

Four creative thinking tasks were applied in the study: two alternative uses tasks (AUT; Torrance, 1966), one consequences task (Torrance, 1966), and one creative problem-solving task. Each creativity task was presented in two steps, where in Step 1, participants engaged in a divergent phase to generate a broad array of ideas. This was followed by a convergent phase (Step 2), during which they were instructed to identify and further elaborate their most promising ideas or develop solutions to the most pressing challenge.

For the AUT, the following instructions were used:

> AUT 1 and 2:
>
> (Step 1): Please list at least 10 different and unusual uses for *PANTS/A FORK*. The goal is to find "creative ideas," i.e., the ideas that strike other people as *clever, unusual, interesting, extraordinary, humorous, innovative, silly*, or *different*.
>
> (Step 2): Please review all the ideas you generated in the last step and select *the 3 most creative ones* with the help of your peer [Google/ChatGPT]. Please enter the three most creative ideas below and, with the support of your partner, write a short text to develop the idea further and make it both *original* and *useful*.

These instructions were adapted from Nusbaum, Silvia, and Beaty (2014). Explicit instructions that emphasize quantity, creativity, and quality have proven to increase creative performance (Acar, Runco, & Park, 2020).

Similar instructions were used for the consequences task:

> Consequences task:
>
> (Step 1): Suppose that people no longer needed or wanted food; what would be the possible consequences? Please list as many consequences as possible. The consequences can be both positive and negative.

The selection of the consequences task for this study was informed by a preliminary pilot study conducted with 22 students (M_{age} = 21.72, *SD* = 1.60) from the same university 2 months before the main study. We introduced five commonly used consequences tasks to the participants, asking them to rank them according to interest and feasibility. Almost half of the participants (47.3%) identified the food task as one of their top preferences.

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In Step 2 of the consequences task, we requested the participants to focus on the most serious consequence and develop an original and useful solution—two defining characteristics of creativity (Runco & Jaeger, 2012).

Creative problem-solving:

> *(Step 2 of the consequences task)*: Please review all the consequences you generated in the previous step and select *the most serious consequence* with the help of your assigned partner [peer student/ Google/ChatGPT]. Please write a short text with the support of your assigned partner [peer student/ Google/ChatGPT] to suggest an *original* and *useful* solution to this problem.

Scoring creativity tasks

Following the literature on creativity scoring (e.g., Reiter-Palmon, Forthmann, & Barbot, 2019; Silvia et al., 2008), we applied a *subjective scoring* method to score the creativity tests. Six business psychology students (four females, two males, M_{age} = 28.33, SD = 5.24) scored the creativity level of all responses using *uncommonness*, *remoteness*, and *cleverness* as indicators of creativity (Silvia et al., 2008). They also scored *creativity* of the most serious consequence of the consequence task and the *effectiveness* of the solution suggested to the identified problem.

The intra-class correlation coefficient (ICC) of the judges' scores for the AUT tasks indicated high reliability, with $ICC_{(3,6)}$ = .83 for the "pants" task and $ICC_{(3,5)}$ = .80 for the "fork" task. The reliability of the scores for the consequences task was similarly high ($ICC_{(3,5)}$ = .83), as were the scores for solution creativity ($ICC_{(3,5)}$ = .66) and solution effectiveness ($ICC_{(3,6)}$ = .87). For the AUT (fork) task, the consequences task, and the solution creativity scores, we omitted the ratings from one judge in each case as it resulted in an improvement in the ratings' reliability.

As an alternative method of estimating creativity of participants' responses, we used the top-scoring method (e.g., Silvia, 2011; Silvia et al., 2008) that averaged across a participant's top two responses.[2] Top-scoring methods are often preferred over average scoring to avoid confounding with fluency and thus enhancing the discriminant validity of this creativity indicator (Benedek, Mühlmann, Jauk, & Neubauer, 2013). At the same time, top-scoring methods might display lower reliability as compared to average scoring (e.g., Silvia, 2011) and result in the loss of information, as only two ideas are scored for each person, leaving the rest unutilized (e.g., Forthmann, Szardenings, & Holling, 2020). To address these limitations, we adopted a recently proposed multidimensional top-scoring method (Forthmann, Karwowski, & Beaty, 2023), which retains information from all responses: those scored highest for creativity and the remaining ones. We used this scoring method to estimate creativity scores in all tasks where multiple responses were generated, that is, AUT (pants), AUT (fork), and consequences task. By doing so, we aimed to ascertain the robustness of our findings despite the operationalization of creativity.

We also scored the participants' responses in creativity tasks using automated scoring with large language models (LLMs), a form of supervised learning where models are trained on past responses rated by human judges. Specifically, we utilized a beta version of a recent multilingual, multitask model, trained on GPT 3.5, from the open creativity scoring with artificial intelligence (Ocsai) platform (https://openscoring.du.edu/; Organisciak et al., 2023). The model used, Ocsai 1.5, was trained on a variety of open datasets, and the scores obtained using the model range from 1 to 5, where 1 is *very unoriginal*, 5 is *very original*, and 3 is the *median*.

Due to the complex nature of the problem-solving task, where each participant generated a solution for a different problem previously stated in the consequences task, the solutions were omitted from the Ocsai scoring. The irregularities in how the problems were formulated and the need to take into account the instructions for the consequences task to comprehend these problems made effective prompting challenging. Thus, our exploratory analyses focused on the Ocsai scores for divergent thinking (DT) task responses.

We conducted the Ocsai scoring procedure twice for each DT task, with prompts written in both English and German. The correlations between the Ocsai scores with human judges' scores were low-to-medium

[2] These are the top two creative ideas rated by the raters, which are not necessarily the same as the top three creative ideas chosen by the participants. The comparison of the average scoring and top 2 scoring by raters is presented in Figure 3 in the results part.

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across the tasks: pants (idea level: r_{ENG} = .45, r_{GER} = .32; task level: r_{ENG} = .64, r_{GER} = .51), fork (idea level: r_{ENG} = .27, r_{GER} = .30; task level: r_{ENG} = .14, r_{GER} = .28), and consequences (idea level: r_{ENG} = .17, r_{GER} = .27; task level: r_{ENG} = .24, r_{GER} = .33). Given the slightly higher correlations obtained using German prompts in two of the three tasks, we used these scores in further analyses for consistency.

RESULTS

The Supplementary Online Material (SOM) Tables S1 and S2 present descriptive statistics and correlations between study variables. The two human–ChatGPT conditions (with basic vs. specific instruction) did not differ significantly in any variable of interest ($p > .05$ in all cases); thus, we combined them into a single human–ChatGPT condition ($N = 81$) for the subsequent analyses. In what follows, we present the findings in three subsections, focusing on differences between conditions in the task and partner perceptions (RQ1), task-induced changes in participants' creative confidence (RQ2), and differences in creative performance (RQ3).

HOW DID PARTICIPANTS PERCEIVE THE COCREATION ACTIVITY ACROSS CONDITIONS (RQ1)?

We began our analyses with between-group comparisons focusing on participants' perceptions of task interest, difficulty, and mental effort invested in its completion. Since task difficulty and mental effort ratings were highly correlated ($r = .65$), we averaged them into a single cognitive load measure, as both are often used to operationalize this construct (Sweller, 2018), also in the creativity literature (de Chantal, Zielińska, Lebuda, & Karwowski, 2023; Redifer, Bae, & Zhao, 2021). Additionally, we looked at the assessments of partner helpfulness in the task performance and participants' contributions to this process.

There were significant and large differences across conditions in the cognitive load imposed by the task, $F(2, 199) = 58.04$, $p < .001$, $\eta^2_p = .37$, and moderate differences in the interest it generated, $F(2, 199) = 6.03$, $p = .003$, $\eta^2_p = .06$. Participants in the human–human and human–Internet conditions found the task similarly difficult and demanding, with their ratings being significantly and substantially higher than those in the human–ChatGPT condition (Figure 1a). Conversely, the task was least interesting to the human–Internet group, while the human–human and human–ChatGPT conditions perceived the cocreation process as equally engaging (Figure 1b).

Participants working with another person, with the aid of the Internet, or with ChatGPT differed in how helpful they found their partner in task performance, with a very large effect size of differences, $F(2, 199) = 78.84$, $p < .001$, $\eta_p^2 = .44$. Both human and ChatGPT partners were perceived as similarly helpful; however, participants utilizing Google Search during the task rated this tool as significantly less helpful compared to the other conditions (Figure 1c). These findings were further elaborated in the estimates of participants' and their partners' contributions to the task. A mixed ANOVA with the condition as a between-subject factor and contributor as a within-subject factor revealed a significant and very large interaction effect, $F(2, 199) = 138.39$, $p < .001$, $\eta_p^2 = .58$. In the human–human pairs, the perceived contribution of both sides was equal, that is, close to 50% (Figure 1d). For the remaining conditions, the patterns were opposite: participants in the human–Internet condition rated their contribution much higher than their partners, while in the human–ChatGPT condition, people attributed most of the initiative to their AI partner (Figure 1d).

HOW DID THE TASK PERFORMANCE AFFECT PARTICIPANTS' CREATIVE CONFIDENCE (RQ2)?

To examine the impact of task performance on participants' creative confidence, we conducted a mixed ANOVA with the condition as a between-subject factor, time (pre-, posttask) as a within-subject factor, and creative confidence as the dependent variable (see Figure 2). We found no significant main effects for condition, $F(2, 199) = 1.88$, $p = .155$, or time, $F(1, 199) = 2.11$, $p = .148$, but there was a significant *Condition* × *Time* interaction, $F(2, 199) = 5.58$, $p = .004$, $\eta_p^2 = .053$. Posthoc tests with Holm's correction indicated that creative confidence significantly increased from pre- to posttask in the human–human group only, $t(81) = 3.63$, $p < .001$, $d = 0.40$. Participants' self-perceptions did not change in the human–Internet, $t(38) = 0.10$, $p = .918$, or in the human–ChatGPT, $t(80) = 0.90$, $p = .373$, groups. Thus, while there were no initial differences in creative confidence across conditions, $F(2, 199) = 0.74$, $p = .479$, posttask assessments differed, $F(2, 199) = 3.40$, $p = .035$, $\eta_p^2 = .033$. Specifically, participants in the human–human condition exhibited higher confidence than the human–ChatGPT group, $t(161) = 2.66$, $p = .025$, $d = 0.42$. The results for the human–Internet condition remained inconclusive, showing no difference from both the human–human, $t(119) = 0.99$, $p = .568$, and human–ChatGPT, $t(118) = 1.08$, $p = .568$, groups in posttask creative confidence.

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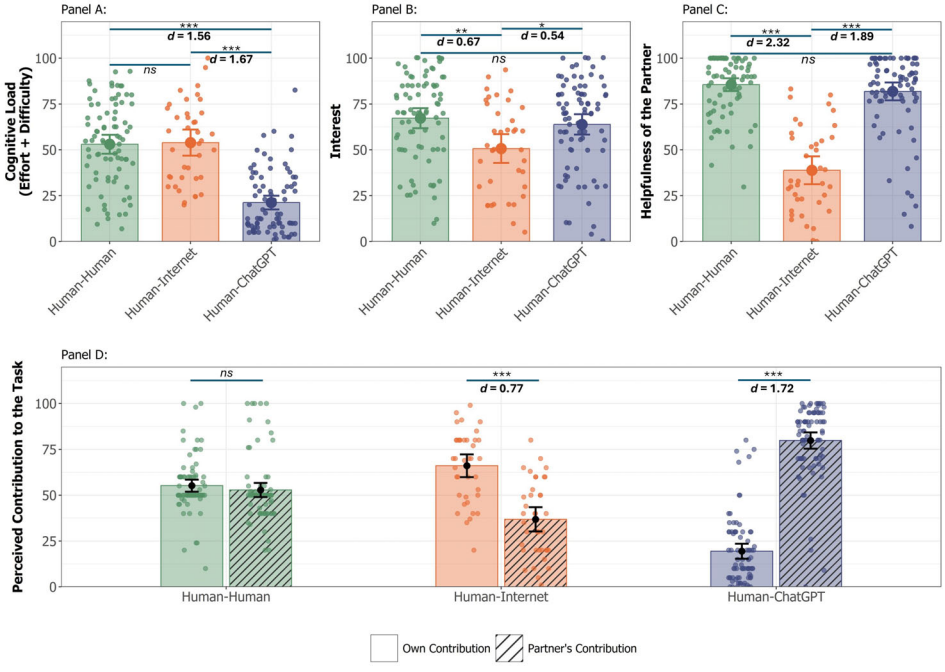


FIGURE 1. Between-group differences in task-imposed cognitive load (a), task interest (b), a partner's helpfulness in the task performance (c), and one's own and partner's contribution to the task (d). *Note*. Holm's correction was applied for the posthoc pairwise comparisons. d = Cohen's d. ns = nonsignificant differences. ***$p < .001$, **$p < .010$, *$p < .050$.

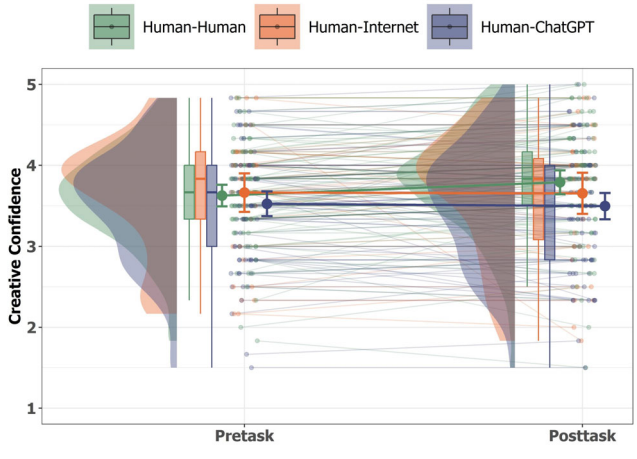


FIGURE 2. Changes in participants' creative confidence beliefs as measured before and after task completion across the study conditions.

HOW DID CREATIVE OUTCOMES DIFFER ACROSS GROUPS (RQ3)?

Next, we examined whether and to what extent the creativity levels of participants' responses differed among conditions. We first reported the results based on human ratings and then compared them with the

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TABLE 3. Factor Loadings of the Overall Divergent Thinking (DT) Performance Factor Obtained in an Exploratory Factor Analysis with Three DT Tasks Included (Separately for Human and Ocsai Scores) and the First, Unrotated Factor Saved

Divergent thinking task	Human ratings (average)	Human ratings (top 2)	Ocsai ratings (average)	Ocsai ratings (top 2)
AUT: Pants	.797	.775	.883	.863
AUT: Fork	.709	.736	.839	.814
Consequences task	.603	.712	.745	.676

Note. AUT = alternate uses task.

AI-based automated scorings. For simplicity, we estimated an overall divergent thinking (DT) performance score through exploratory factor analysis (EFA), retaining a single factor from participants' scores across three tasks: "pants," "fork," and consequences (see Table 3 for factor loadings). We employed a similar approach for the top 2 scores obtained for these tasks. For transparency, Figures S1–S3 in SOM present the results from between-group comparisons run independently for each task.

There were significant differences in overall DT creative performance across conditions, quantified with both average, $F(2, 146) = 5.79$, $p = .004$, $\eta_p^2 = .07$ (medium effect) and top 2: $F(2, 146) = 12.80$, $p < .001$, $\eta_p^2 = .15$ (large effect) creativity ratings. Posthoc tests further revealed that the human–human condition outperformed the human–Internet and human–ChatGPT groups, demonstrating moderate-to-large effect sizes (Figure 3). We emphasize that these effects were driven mainly by the performance in the "fork" task,

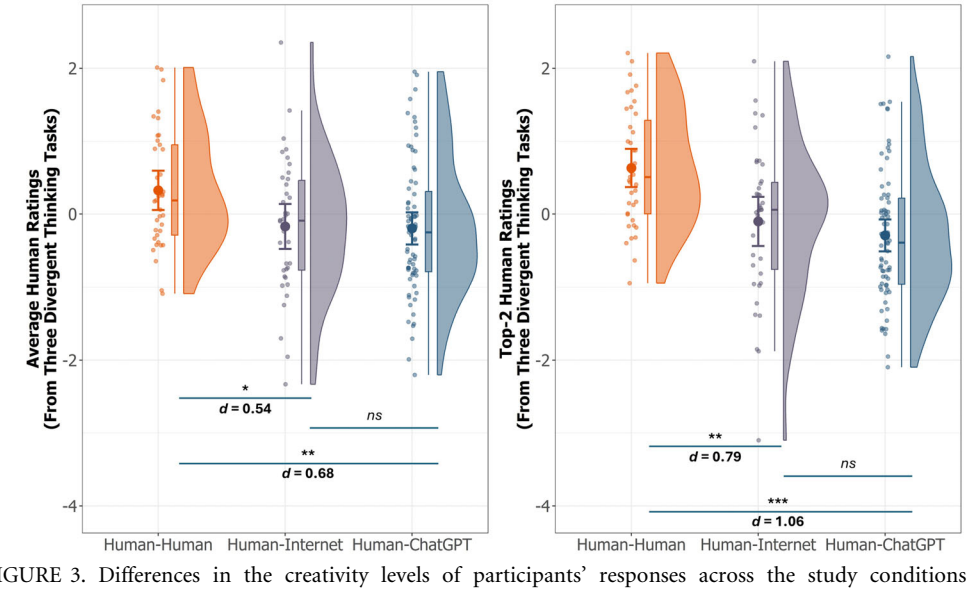


FIGURE 3. Differences in the creativity levels of participants' responses across the study conditions as obtained using two scoring methods: average scoring (left panel), and top 2 scoring (weighted most creative responses; right panel). *Note*. Divergent thinking (DT) scores were derived from exploratory factor analyses (EFAs), which retained single factors (factor score variables are standardized, $M = 0$, $SD = 1$) based on participants' scores across three tasks: "pants," "fork," and consequences. The average human ratings index uses average scores from each task, while the top 2 human ratings index is based on the top 2 scores obtained for these tasks. Holm's correction was applied for the posthoc pairwise comparisons. d = Cohen's d. ns = nonsignificant differences. ***$p < .001$, **$p < .010$, *$p < .050$.

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as differences in the other two tasks were negligible (see SOM Figures S1–S3). The human–Internet and human–ChatGPT conditions did not differ significantly in overall DT creative performance (Figure 3).

These patterns replicated when focusing on responses that participants chose as their best. We calculated the overall creativity score for self-selected ideas first by averaging ratings for three chosen ideas for each task separately (i.e., "pants" and "fork"; note that for the consequences task, the participants were asked to pick the most severe consequence, not the most creative one; thus, the consequence task was excluded from the overall creativity score for self-selected ideas), and then averaging these task-level scores. The groups differed in the creativity score for self-selected ideas, $F(2, 158) = 5.73$, $p = .004$, $\eta_p^2 = .07$, with participants from the human–human condition scoring substantially higher than human–ChatGPT condition, $t(120) = 3.33$, $p = .003$, $d = 0.60$, and marginally higher than human–Internet condition, $t(78) = 1.85$, $p = .086$, $d = 0.41$. There were no differences between human–Internet and human–ChatGPT conditions, $t(118) = 1.09$, $p = .326$.

Finally, we observed no differences between the groups in the problem-solving task. Both the creativity, $F(2, 158) = 0.32$, $p = .730$, and effectiveness, $F(2,158) = 1.13$, $p = .326$, scores for the solutions proposed were similar across all conditions.

DO THE RESULTS REPLICATE WITH RESPONSES SCORED WITH AI-BASED TOOLS?

To further explore the consistency of our findings, we tested whether the observed between-group differences in creative performance would replicate when responses are scored using Ocsai, an LLM-based scoring system. As with the human judges' ratings, we calculated an overall DT performance score by retaining a single factor from three DT tasks in an EFA (see Table 3 for factor loadings). The correlations between the DT score based on human ratings and the Ocsai scores were moderate for both the average ($r = .36$) and top 2 scores ($r = .25$), and lower for the scores averaged across participants' self-selected ideas ($r = .20$).

Consistent with previous findings, overall DT creative performance differences across conditions were significant when considering average scores, $F(2, 146) = 8.35$, $p < .001$, $\eta_p^2 = .10$, top 2 scores, $F(2, 146) = 7.01$, $p = .001$, $\eta_p^2 = .09$, and participants' self-selected ideas, $F(2, 158) = 6.99$, $p = .001$, $\eta_p^2 = .08$, were considered. Surprisingly, however, the direction of these differences was reversed compared to the pattern observed with human judges' scores. Posthoc tests showed that it was the human–ChatGPT groups that outperformed both the human–human (average scores: $t(108) = 3.57$, $p = .002$, $d = 0.78$; top 2 scores: $t(108) = 3.18$, $p = .006$, $d = 0.68$; self-selected ideas: $t(120) = 3.06$, $p = .008$, $d = 0.64$) and human–Internet conditions (average scores: $t(108) = 2.41$, $p = .035$, $d = 0.51$; top 2 scores: $t(108) = 2.55$, $p = .024$, $d = 0.53$; self-selected ideas: $t(118) = 2.51$, $p = .027$, $d = 0.53$), with moderate-to-large effect sizes. The human–human and human–Internet conditions did not differ significantly from each other (average scores: $t(76) = 1.52$, $p = .132$; top 2 scores: $t(76) = 0.63$, $p = .530$; self-selected ideas: $t(78) = 0.66$, $p = .509$).[3]

To shed light on the potential reasons behind these inconsistencies, we examined whether the between-condition differences held after controlling for the elaboration of participants' ideas. Elaboration was operationalized as the number of characters in the responses provided across the tasks (square rooted for each task due to skewed distribution, then aggregated). This additional analysis was inspired by the observation that the conditions indeed differed in elaboration, $F(2, 146) = 4.90$, $p = .009$, $\eta_p^2 = .06$, with the human–ChatGPT groups providing more elaborate responses than the human–human condition, $t(108) = 2.66$, $p = .027$, $d = 0.59$, but not human–Internet group, $t(108) = 1.74$, $p = .168$. The differences between human–human and human–Internet groups in elaboration were insignificant, $t(76) = 1.75$, $p = .168$. Furthermore, while elaboration was only weakly related to human ratings ($r = .20$), it was very highly correlated with Ocsai scores ($r = .82$). Thus, we hypothesized that the reversed pattern of between-group differences observed when analyzing automated scores might be due to elaboration bias and Ocsai tends to score more elaborate responses higher.

[3] At the time this article was revised, Ocsai had been updated to version 1.6, which was trained on GPT-4.0 instead of GPT-3.5. Thus, we examined whether our results remained consistent with this new version, focusing on average scores (as before, we used an EFA based on the scores from three DT tasks to obtain an indicator of overall DT performance). The correlation between Ocsai scores obtained using the previous and updated versions was high for overall DT performance ($r = .87$), suggesting strong similarity, though the scores were not identical. However, the updated DT score exhibited nearly identical correlations with human ratings ($r = .37$) as Ocsai 1.5 ($r = .36$). Importantly, the pattern of between-condition differences was exactly the same as with the Ocsai 1.5 scores, with even more robust effect sizes. The omnibus effect of condition was significant, $F(2, 146) = 12.53$, $p < .001$, $\eta_p^2 = .15$, and further posthoc comparisons indicated that the human–ChatGPT group outperformed both the human–human, $t(108) = 4.79$, $p < .001$, $d = 1.01$, and human–Internet conditions, $t(108) = 2.43$, $p = .033$, $d = 0.50$.

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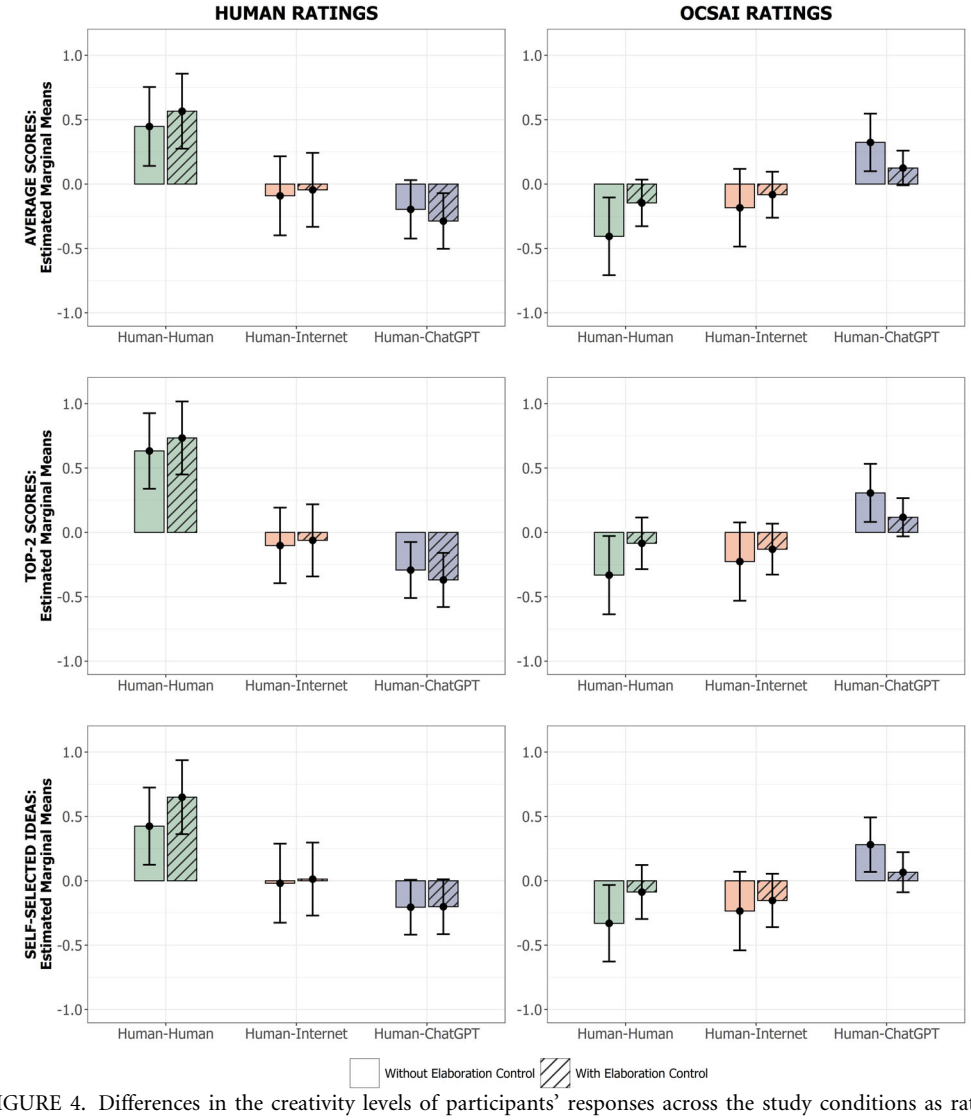


FIGURE 4. Differences in the creativity levels of participants' responses across the study conditions as rated by human judges (left panel) or Ocsai (right panel), with and without elaboration as a covariate. *Note*. Overall average and top 2 scores were calculated using exploratory factor analyses (EFAs), which identified single factors (factor score variables were standardized, $M = 0$, $SD = 1$) based on participants' performance across three tasks: "pants," "fork," and "consequences." The "average scores" index reflects the mean scores from each task, while the top 2 human ratings index is based on the top two scores obtained across these tasks. The index for self-selected ideas was computed as the average rating of the ideas that participants themselves identified as their best in two tasks ("pants" and "fork") and is presented in standardized form for comparability.

Indeed, when controlling for elaboration in predicting Ocsai ratings in ANCOVA, the effects of condition were weakened for average scores, $F(2, 145) = 3.27$, $p = .041$, $\eta_p^2 = .04$, and even disappeared for the top 2 scores, $F(2, 145) = 2.34$, $p = .100$, and self-selected ideas, $F(2, 145) = 1.54$, $p = .218$ (see Figure 4). In

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contrast, when elaboration was included as a covariate in the comparisons based on human ratings, the previously observed effects became even more pronounced (see Figure 4). Specifically, the differences between groups remained significant—and in a direction opposite to that indicated by Ocsai ratings—for average scores, $F(2, 145) = 10.60$, $p < .001$, $\eta_p^2 = .13$, top 2 scores, $F(2, 145) = 18.73$, $p < .001$, $\eta_p^2 = .21$, and self-selected ideas, $F(2, 145) = 10.93$, $p < .001$, $\eta_p^2 = .13$.

In sum, the relatively weak associations between human judges' and Ocsai scores of response creativity (*r*s between .20 and .36),[4] alongside the strong correlation between Ocsai scores and elaboration ($r = .82$), and the diminishing or even disappearing differences between human–human and human–GPT conditions in Ocsai scores after controlling for elaboration, lead us to consider human judges' scores as more credible and valid. The reversed pattern of differences depending on how the responses are scored is intriguing, as it suggests that Ocsai (at least in German) might overestimate creativity scores when responses are longer or provided with the help of AI, yet for the sake of our comparisons, the classic human judges' scores are more trustworthy.

DISCUSSION

Although AI has been around for quite some time, the release and broad availability of ChatGPT at the end of 2022 made its use for various purposes—including scholarly ones—more prevalent than ever. After the initial hype, 2 years of living with constantly improving versions of generative AI made the benefits and limitations of AI-based solutions better understood and empirically tested. Consequently, while the initial public discourse viewed generative AI as a revolutionary solution that will replace humans in the job market, particularly in complex, creative jobs, the current status quo calls for a more nuanced perspective regarding its potential and limitations.

In creativity research, AI-based solutions have proven useful in streamlining the scoring of various creative thinking tests, making the research process more efficient. Indeed, solutions such as *Ocsai* (Organisciak et al., 2023; Zielińska et al., 2023) or an automated drawing assessment platform for scoring figural creativity tasks (AuDrA; Patterson et al., 2024) are not only fast, but also the scores are often indistinguishable from human ratings. The rapid development of automated scoring of short stories (Johnson et al., 2022; Luchini et al., 2024), metaphors (DiStefano et al., 2024), and problem-solving solutions (Luchini et al., 2023), as well as a recently created creativity test for children entirely based on automated scoring (Acar et al., 2024) would not have been possible without the opportunities that generative AI has created. Additionally, AI's potential for developing or elaborating new psychometric measures is unprecedented (Beghetto, Ross, Glaveanu, & Karwowski, 2025; Glăveanu, Karwowski, Ross, & Beghetto, 2024).

Yet, while the usefulness of AI for making creativity research more efficient is largely uncontroversial, it is more difficult to unanimously answer the question of the role AI might play as a tool to enhance people's creative thinking. A recent manifesto (Vinchon et al., 2023) considered various roles of AI in and for creativity: from being an external tool used by humans to solve problems and generate ideas to acting as an equal-right collaborator, all the way to seeing it as an agentic entity capable of being creative without human's input. However, the current state of generative AI makes considering it as a creative agent the least convincing, as AI still lacks the intention to behave creatively, a critical factor for creative agency (Beghetto & Karwowski, in press; Karwowski & Beghetto, 2019). Yet, considering generative AI as a tool or a partner is much more reasonable.

The current experiment explored the process and outcomes of collaborative interactions between humans and AI. Unlike previous studies that primarily compared human- and GPT-generated solutions in various creativity tasks (mostly divergent thinking tasks, Hubert et al., 2024; Stevenson, Smal, Baas, Grasman, & van der Maas, 2022; or problem-solving, Luchini et al., 2023; but also argumentative essays, Herbold, Hautli-Janisz, Heuer, Kikteva, & Trautsch, 2023), we were interested in comparing various forms of cooperation: human–human, human–Internet (Google Search), and human–ChatGPT. Our reasoning was that relying on a simple comparison of ideas generated by humans and AI might be inconclusive, given the massive databases large language models were trained on and are available to AI, while obviously unavailable to

[4] Not only are those correlations far lower than reported in other recent studies with the use of Ocsai (Organisciak et al., 2023), including scoring responses in languages other than English (Zielińska, Lebuda, Gop, & Karwowski, 2024), but it is also far below what was observed in Ocsai training data for tasks in various languages, German included. Ocsai 1.6 (Peter Organisciak, personal communication, August 23, 2024, see also https://buttondown.com/creativity/archive/ocsai-16-an-easy-swap-in-update/) estimated correlations between human judges and AI estimates of alternate uses task vary between $r = .45$ for Arabic and $r = .81$ for German, with the validity for German being even higher than the correlation for English ($r = .76$).

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individual participants. Thus, we complemented our design with an additional group: participants cooperating with the Google Search engine, who could search for and find potential solutions that were inaccessible while working alone. Given that human–Internet (Google Search) and human–AI (ChatGPT) groups involved collaborations between humans and external enhancement tools, a proper comparison group was not individual participants but human–human dyads, allowing participants to exchange their perspectives and share information and knowledge.

Our study involved various tasks and aspects relevant to the cocreation process. Thus, we not only included different creativity measures: two AUT tasks, a consequence task, and a problem-solving task, but also analyzed changes in participants' creative confidence, their perception of cognitive load and interest during the process, and the perceived contribution and helpfulness of the "partner" they collaborated with. Next, we discuss the main findings from the current investigation.

WORKING WITH HUMANS VERSUS WORKING WITH AI: *DIFFERENCES IN THE PERCEIVED PHENOMENOLOGY OF THE PROCESS*

Probably the most prevalent set of our findings—with large effect sizes of the differences observed—relates to our first research question (RQ1), that is, participants' perception of the various aspects of collaborations we initiated in the study. People working with others (human–human condition) or using Google to create their responses found the whole process much more difficult and effortful (imposing a higher cognitive load) than those in the human–AI condition. The differences were substantial (both *d*s >1.50). Thus, using AI makes the process easier, at least as perceived by participants, while not diminishing the level of interest. When comparing the level of interest across conditions, the only significant differences were between human–human versus human–Internet groups ($d = 0.67$) and human–AI versus human–Internet groups ($d = 0.54$). In other words, searching for solutions on Google was perceived as much less engaging and more difficult. This perception could be attributed to the comparatively poor user experience offered by traditional search engines like Google Search when compared to large language models (LLMs) based models such as ChatGPT (Marr, 2023), which has spurred discussions about the inevitable evolution of traditional search engines into "thinking" machines through the integration of search functionality and the LLMs (Goodwin, 2024; Marr, 2023).

The largest observed differences between groups were related to the perceived helpfulness of the "partner." People working with another person or ChatGPT saw their partner as extremely helpful. Accordingly, both conditions showed higher perceived partner's helpfulness than the human–Internet condition ($d = 2.32$ for a human–human versus human–Internet comparison and $d = 1.89$ for human–AI versus human–Internet conditions). Finally, we observed a much-telling interaction when participants were asked about their own (actor) and their partner's contributions. In the human–human condition, the perceived contribution of both persons in dyads was almost the same (around 50%); in the human–Internet condition, people perceived their role as much more dominant than their "partner's" ($d = 0.77$), while in the human–AI condition, the opposite effect was observed with a large effect size: here ChatGPT was clearly leading ($d = 1.72$).

Although based on declarative data, the observed patterns speak for our investigation's internal validity and provide potential explanations for the effects of participants' creative self-perceptions. If people perceive AI (ChatGPT) as the main contributor to the creative process, how might this influence their creative confidence? Working with AI is perceived as easier (compared to other conditions), but does being easy equal being creative? We explore these questions in the remaining part of our Discussion.

WORKING WITH HUMANS VERSUS WORKING WITH AI: *EFFECTS ON CREATIVE CONFIDENCE*

Creative confidence is a vital motivational factor that explains people's engagement or lack of engagement in creative activities. If people are unconvinced about their creative potential, they see little reason to engage in ill-defined tasks. And quite the opposite: those more confident are not only more successful in attaining creative outcomes (Karwowski & Beghetto, 2019; Tierney & Farmer, 2011), but also more effective in managing the process (Zielińska et al., 2023, 2024).

In our study, we measured participants' creative confidence twice: before engaging in collaborative creativity and at the end of this process. As expected, we found high stability in trait-like creative confidence (see Beghetto & Karwowski, 2017; Karwowski, Han, & Beghetto, 2019; Zandi, Karwowski, Forthmann, & Holling, 2022) with a strong correlation between the first and second measurements across all conditions, $r = .84$ ($p < .001$). Still, while searching for an answer to our second research question (RQ2), we observed

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a significant interaction effect with stability varying depending on the conditions. In particular, while participants' initial creative confidence levels were similar across conditions (attesting to adequate randomization), there was a significant and nontrivial (d = 0.40) increase in creative confidence among participants in the human–human condition. No such changes were found in the remaining conditions, where participants worked with either Google Search or ChatGPT.

Although preliminary and requiring replication and investigation into the exact mechanisms, the increase in creative confidence might be theoretically and practically sound. Creative confidence is a subset of creative agency (Karwowski & Beghetto, 2019), so observing its growth in dyadic collaborations is promising, as it suggests that such interactions can be a way to enhance it. Furthermore, this opens up new avenues to explore possible overlooked paths to theorize how creative agency develops in dyads, teams, and groups. Moreover, linking this increase to previously observed differences in the perceived phenomenology of the process seems even more interesting. Recall that participants' creative confidence increased only in a particular condition: the one perceived as both challenging and interesting, with a highly helpful partner who did not dominate the process (i.e., the perceived contribution of an actor and the partner was similar). Such a combination of challenge and interest, with constructive cooperation, resembles the definitional conditions of the flow state (Csikszentmihalyi, 1990), in our case, dyadic flow, which seems to contribute to enhanced participants' creative confidence. Of course, we emphasize the speculative nature of this observation. Future research should explore the interactions within human–human dyads in more detail (e.g., Glăveanu, Gillespie, & Karwowski, 2019) to fully untangle the processes behind this pattern.

Given the trait-like and domain-general character of our measurement of creative confidence (Beghetto & Karwowski, 2017), some more dynamic effects associated with task-specific creative self-efficacy might have been obscured. For example, a recent study (Urban et al., 2024) showed that people solving creative problems with the help of ChatGPT hold higher task-specific self-efficacy than those working without AI, meaning they felt more confident in their ability to propose original and creative solutions when being augmented by AI. Interestingly, however, this study found it unlikely that perceived task-related self-efficacy was responsible for strengthening the generated solutions' originality. Although the group working with ChatGPT provided significantly more original responses compared to the baseline situation (and outperformed the originality of solutions generated by participants working alone), this change was not due to higher task-specific self-efficacy. Nevertheless, since that study did not focus on human–human collaboration, it is not directly comparable to our findings. Therefore, future studies would benefit from incorporating both trait-like and task-specific creative confidence measures as potential drivers of changes in originality. This further applies to the potential shifts in AI-specific creative self-perceptions (Faiella et al., 2024), which are shaped specifically within the context of human–AI collaboration.

WORKING WITH HUMANS VERSUS WORKING WITH AI: *EFFECTS ON CREATIVE OUTCOMES*

So far, we have focused on differences in the phenomenology of the process, that is, its perceived character across conditions, as well as the promise of human–human collaboration to build creative confidence. Yet, what are the effects of various forms of collaboration on the outcomes achieved? That was precisely the focus of our third research question (RQ3). While using human judges, who scored participants' responses with high reliability—a gold standard in creativity research (Long & Plucker, 2015; Plucker & Makel, 2010)—we also applied Ocsai, an AI-based scoring system, as an independent check to see if the results would replicate. As it became apparent, they did not.

Regarding human judge-provided scores—our primary operationalization of solutions' originality—the overall pattern obtained was stable despite the specific originality scoring variant (i.e., average scoring, top 2 scoring, participants-selected scoring). Human–human dyads generally outperformed human–Internet dyads and, especially, human–AI dyads. The effect sizes of the differences between human–human and human–AI varied from moderate (d = 0.60 and d = 0.68 when participants-selected ideas scoring and average scoring were applied, respectively) to large (d = 1.06) when top 2 scoring was used. In none of the cases did the human–Internet and human–AI conditions differ in creativity.

Thus, when human judges scores were used, the pattern of differences we observed was robust despite the way of estimating originality, showing the advantage of human–human collaboration over working in human–machine dyads. This is intriguing, given previous studies demonstrating that ChatGPT often creates more original responses than humans (Herbold et al., 2023; Hubert et al., 2024; Stevenson et al., 2022). Still, it is consistent with investigations suggesting that ChatGPT might be conducive to productivity but not necessarily idea originality (Boussioux et al., 2024; Dell'Acqua et al., 2023; Doshi & Hauser, 2024). Additionally,

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some studies caution that ChatGPT might reduce problem-solving effectiveness because of its "hallucinated" responses (e.g., Kim & Moon, 2024). Most importantly, our results are not directly comparable to studies with individuals, as our conditions involved dyads: either people working together or individuals working with machines. Thus, given the design, we cannot decisively conclude whether the advantage of the human–human condition stems from ideational elements (e.g., potentially higher flexibility of people working together thanks to perspective taking), a better combination of self- and other-referenced creativity (Dumas, Forthmann, & Alexander, 2024) or perhaps from collaborative evaluation (e.g., people in dyads tend to be more realistically oriented; Glăveanu et al., 2019).

Similarly, the different patterns of the phenomenology of the process identified (the "flow-like" collaborative process in the case of human–human dyads) and the gains in creative confidence we observed may also help explain the differences in outcomes. However, the between-group design of our experiment is not sufficiently powered to test for mediating and moderating effects. Thus, well-powered replications and extensions of this study's findings are crucial.

Our secondary analyses involved scoring the responses using the AI-based tool, Ocsai (Organisciak et al., 2023). We have to admit that the results obtained were pretty puzzling. Although automated creativity scoring systems typically behave excellently—providing estimates that are hardly distinguishable from those by humans—this was not the case here. Despite multiple checks, various models, and both original German responses and their English translations, the correlations between Ocsai estimates and human judges' scores were moderate at best (*r*s about .30). Even more concerning, the pattern we observed based on human scoring was reversed when Ocsai scores were compared across conditions. When correlations are mediocre and human judges' reliability is high, priority is generally given to human ratings—after all, AI scoring is validated against human judgments. Moreover, when we added responses elaboration to the equation, nothing changed in the human judge-based differences, while the Ocsai-obtained differences either disappeared or were reduced by more than half (as evidenced by the change of partial eta-square from 10% in the initial analysis to 4% when elaboration was taken into consideration). Thus, although there is a compelling case that AI-based automated scoring might be more prone to elaboration bias than human judges, the intriguing pattern we observed opens an avenue for future studies—a perspective we briefly discuss next.

STRENGTHS, LIMITATIONS, AND FUTURE DIRECTIONS

This investigation's findings should be read in light of this study's strengths and limitations. Among the strengths, we see multistage, intensive, and quite dynamic experimental design with several creativity measures applied, which allowed for a more robust test of the generalizability of the effects obtained. We also captured participants' perceptions (phenomenology) of the collaborative cocreation process and monitored changes in their creative confidence.

Still, there are limitations worth addressing in future investigations. The first is the lack of differences between the groups when considering their performance in the creative problem-solving task. While our findings—when the tasks were scored by human judges—clearly show higher creativity in divergent thinking tasks when people work with a human partner compared to a machine partner, we cannot conclude that this advantage also extends to problem-solving tasks requiring convergence on a solution. Given that the type of problem influences creativity in creative problem-solving (Reiter-Palmon, Illies, Kobe Cross, Buboltz, & Nimps, 2009), the "nested" design of this task in our study—where participants generated solutions to a problem they had also come up with and selected—makes the process highly idiosyncratic and hardly comparable across groups or dyads. Therefore, future studies should investigate whether and how various cooperative setups affect creative outcomes and processes in more standardized creative problem-solving settings.

However, the most critical limitation of this experiment is our inability to provide an entirely compelling explanation of the effects we observed in terms of psychological or social mechanisms behind the differences in DT performance. For example, we did not record how participants handled unsatisfactory responses from their creative thinking partners (see Kim & Moon, 2024). Although we offered some initial hypotheses about potential mechanisms at play, we did not test them due to insufficient statistical power or lack of the necessary data. Therefore, it is important to combine future experiments with more qualitative, ethnographic studies that will bring new evidence regarding how people interact with ChatGPT, what kind of prompts they are using, and how they select or reject the ideas suggested by AI. Another natural extension is to embed and record the human–ChatGPT interactions into surveys (e.g., Qualtrics), allowing for future analysis of the prompts, exchanges between partners, and the final

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responses provided. Finally, more dynamic designs, potentially including eye-tracking methodologies and tracking passive data, would be particularly beneficial to better understand the process of creative collaboration, either with humans or machines.

As a side note and inspiration for future studies, we want to highlight the puzzling inconsistencies between human judges' and Ocsai's scores. Although conclusive evidence of why Ocsai failed in our case is lacking, a tempting hypothesis is that AI might somehow recognize and favorably evaluate AI-generated responses. If this is true, it would have far-reaching substantial and methodological implications. Substantially, exploring if this is indeed the case could enhance our understanding of how AI models assess creativity and what biases they may be prone to. Methodologically, the consequences are equally vital. In the era of online studies, which are increasingly accompanied by well-known problems with bots (Douglas, Ewell, & Brauer, 2023; Webb & Tangney, 2024), ChatGPT-provided responses to open-ended questions in questionnaires are becoming a growing concern. Suppose automated scoring systems consistently rate ChatGPT-generated responses as more creative than they actually are, as we observed in this study, this could introduce error variance into the results, making the literature noisy. We, therefore, invite fellow researchers to take a closer look at this problem before it becomes too prevalent.

CONCLUSION

This multistage experiment demonstrates that collaborative interaction in the human–human dyads when compared with human–Internet or human–AI collaboration results in more creative outcomes across various divergent thinking tasks. Importantly, working with another human not only boosts creativity but also strengthens a critical agentic element of the self: creative confidence. This study challenges the prevailing assumption that AI surpasses human collaboration in fueling creativity. Instead, it supports the view that creativity is a unique human endowment that is not easily replicated by AI (e.g., Brainard, 2023; Lee & Chen, 2021). Additionally, discrepancies between human and AI scoring systems suggest that AI may be susceptible to elaboration bias, where longer or more detailed responses are overvalued. Future research should explore the specific social and cognitive mechanisms that drive the effectiveness of human–human or human–AI partnerships in creative endeavors, while addressing potential biases in AI scoring systems. Such insights will enable us to value and enhance our human creative potential rather than diminishing it through an overreliance on AI.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

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Table S1. Descriptive statistics and pearson's r correlations among the study variables.
Table S2. Descriptive statistics for the key study variables across the four Cconditions: Human-Human, Human-Internet, Human-ChatGPT (basic instruction), and Human-ChatGPT (specific instruction).
Figure S1. Differences in the creativity levels of participants' responses across the tasks and study conditions as obtained using average scoring method.
Figure S2. Differences in the creativity levels of participants' responses across the tasks and study conditions as obtained using top 2 scoring method.
Figure S3. Differences in the creativity levels of responses chosen by participants as their best across the tasks and study conditions.

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